

7. SIGNAL RECONSTRUCTION AND SPECTRAL ANALYSIS

Experiment 7.1

V

% Reconstruction of Signal from Samples

clc;

clear;

close all;

% Original signal frequency

fm = 5; % Hz

% Sampling frequency

fs = 25; % Hz

% Continuous-time signal

t = 0:0.001:1;

x = sin(2*pi*fm*t);

% Sampling

ts = 0:1/fs:1;

xs = sin(2*pi*fm*ts);

% Reconstruction using interpolation

xr = interp1(ts,xs,t,'spline');

% Plotting

figure;

subplot(3,1,1);

plot(t,x,'LineWidth',2);

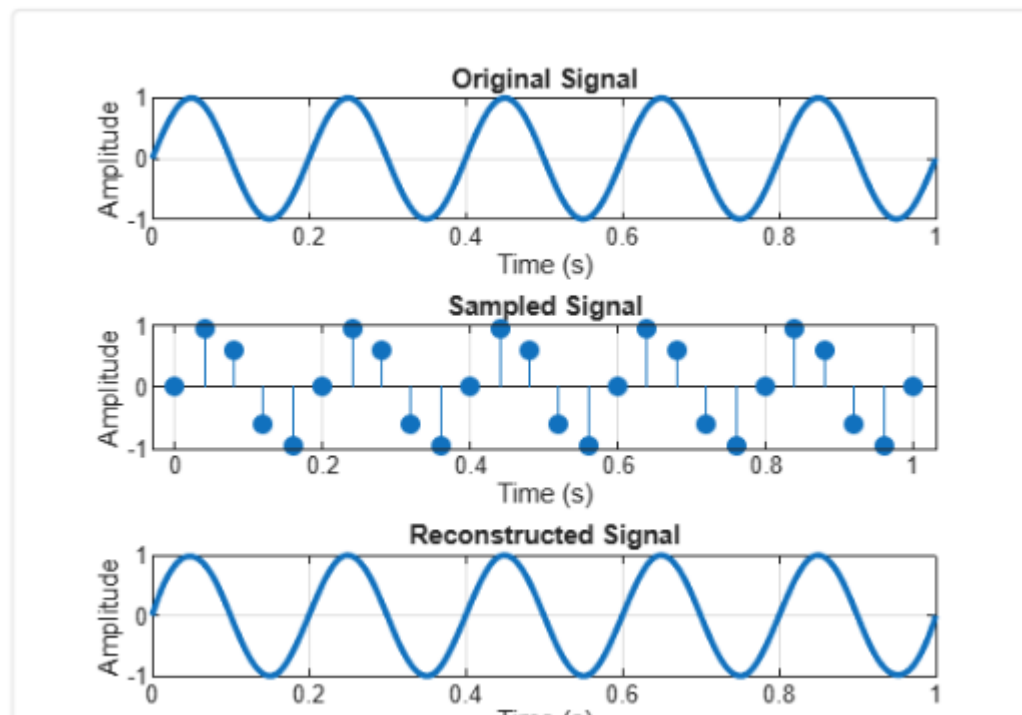
grid on;

title('Original Signal');

```

xlabel('Time (s)');
ylabel('Amplitude');
subplot(3,1,2);
stem(ts,xs,'filled');
grid on;
title('Sampled Signal');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(3,1,3);
plot(t,xr,'LineWidth',2);
grid on;
title('Reconstructed Signal');
xlabel('Time (s)');
ylabel('Amplitude');

```



Experiment 7.2% Fourier Series of Half-Wave Rectifier Output

```
clc;
```

```

clear;

close all;

% Time axis
t = 0:0.001:2*pi;

% Half-wave rectified signal
x = sin(t);
x(x < 0) = 0;

% Fourier Series approximation
N = 10; % Number of harmonics
a0 = (1/pi)*trapz(t,x);
fs_signal = a0/2;
for n = 1:N
    an = (1/pi)*trapz(t,x.*cos(n*t));
    bn = (1/pi)*trapz(t,x.*sin(n*t));
    fs_signal = fs_signal + an*cos(n*t) + bn*sin(n*t);
end

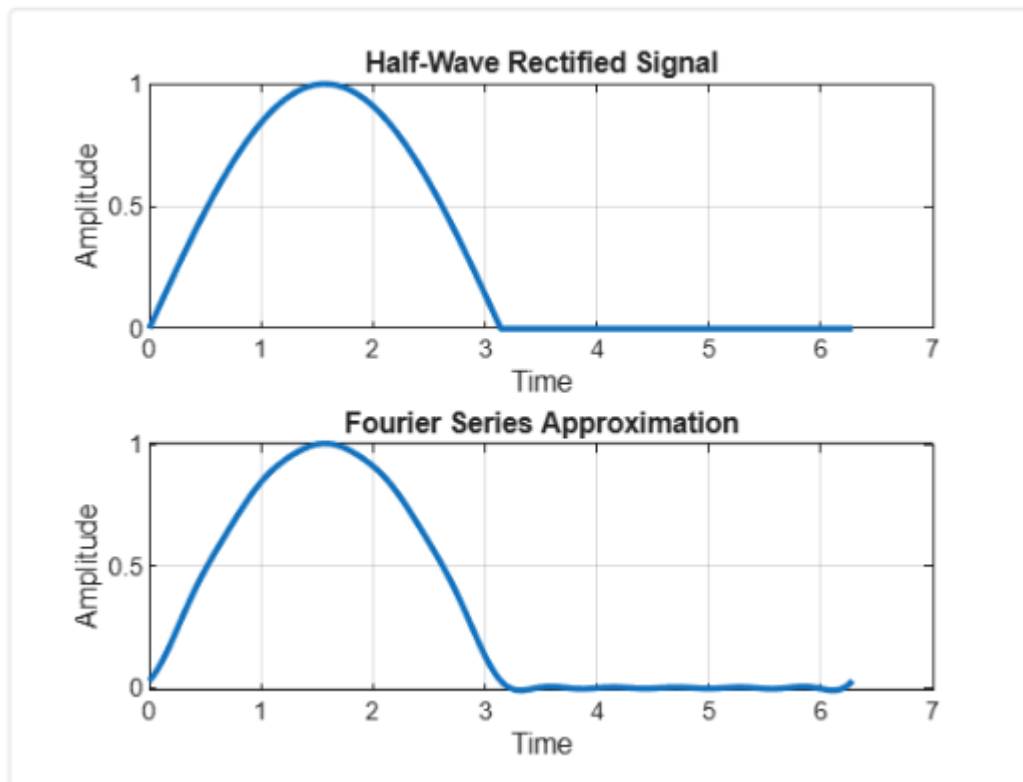
% Plotting
figure;
subplot(2,1,1);
plot(t,x,'LineWidth',2);
grid on;
title('Half-Wave Rectified Signal');
xlabel('Time');
ylabel('Amplitude');
subplot(2,1,2);

```

```

plot(t,fs_signal,'LineWidth',2);
grid on;
title('Fourier Series Approximation');
xlabel('Time');
ylabel('Amplitude');

```



Experiment 7.3vv% Fourier Transform of a Continuous-Time Signal

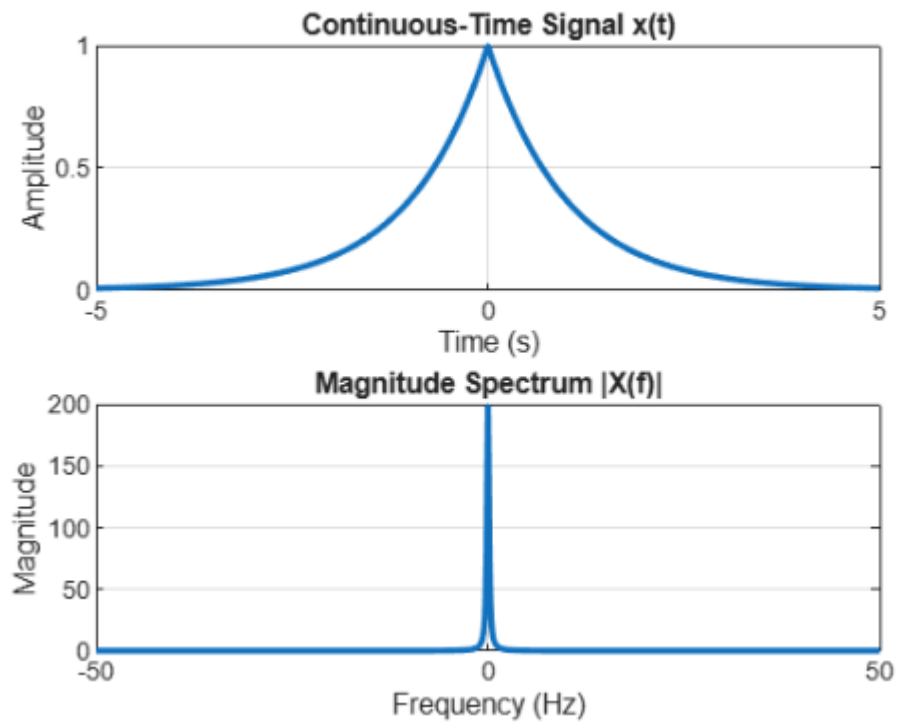
```

clc;
clear;
close all;
% Time axis
t = -5:0.01:5;
% Continuous-time signal
x = exp(-abs(t));
% Fourier Transform using FFT

```

```
X = fftshift(fft(x));  
% Frequency axis  
N = length(X);  
f = linspace(-50,50,N);  
% Plotting  
figure;  
subplot(2,1,1);  
plot(t,x,'LineWidth',2);  
grid on;  
title('Continuous-Time Signal x(t)');  
xlabel('Time (s)');  
ylabel('Amplitude');  
subplot(2,1,2);  
plot(f,abs(X),'LineWidth',2);  
grid on;  
title('Magnitude Spectrum |X(f)|');  
xlabel('Frequency (Hz)');
```

`ylabel('Magnitude');`



% Fourier Transform of a Rectangular Waveform

```
clc;
```

```
clear;
```

```
close all;
```

```
% Time axis
```

```
t = -5:0.01:5;
```

```
% Rectangular pulse
```

```
x = double(abs(t) <= 1);
```

```
% Fourier Transform
```

```
X = fftshift(fft(x));
```

```
% Frequency axis
```

```
N = length(X);
```

```
f = linspace(-50,50,N);
```

```
% Plotting
```

```
figure;
```

```
subplot(2,1,1);
```

```
plot(t,x,'LineWidth',2);  
grid on;  
title('Rectangular Waveform');  
xlabel('Time (s)');  
ylabel('Amplitude');  
subplot(2,1,2);  
plot(f,abs(X),'LineWidth',2);  
grid on;  
title('Magnitude Spectrum');  
xlabel('Frequency (Hz)');  
ylabel('|X(f)|');
```

Experiment 7.4

